



Solar Generation Analysis Dashboard

Solar ♦ Battery ♦ Weather Analytics

 Overview

 Solar Performance

 Battery Analysis

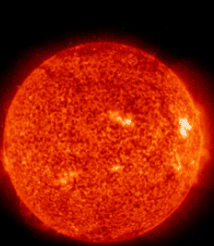
 Weather Analysis

 Grid Analysis

 Hourly Analysis

 Key Influencer

 Conclusion



date

01-02-202601-06-2026

weather_code

All

is_day

All

4.01K

 Total Generation

307.68

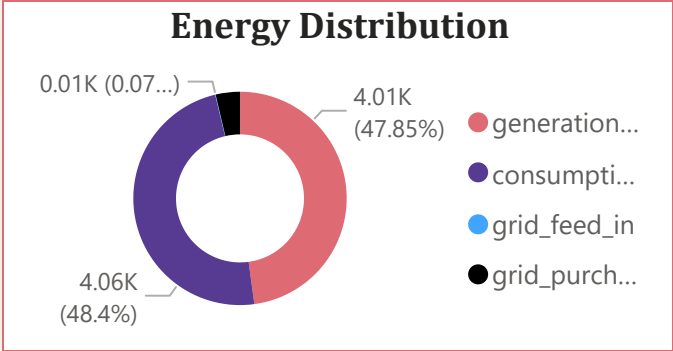
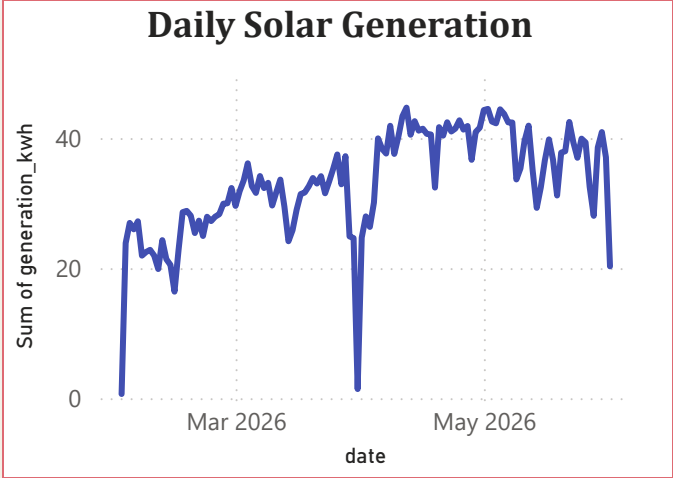
 Grid Purchase

2.00K

 Battery charge

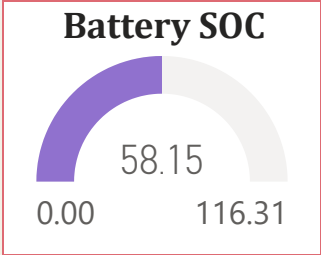
1.80K

 Battery Discharge



6.20

 Grid Feed In

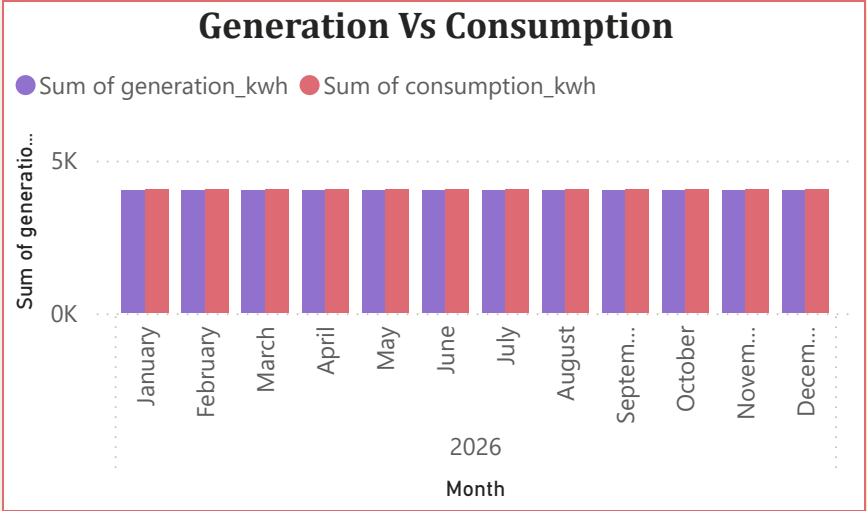
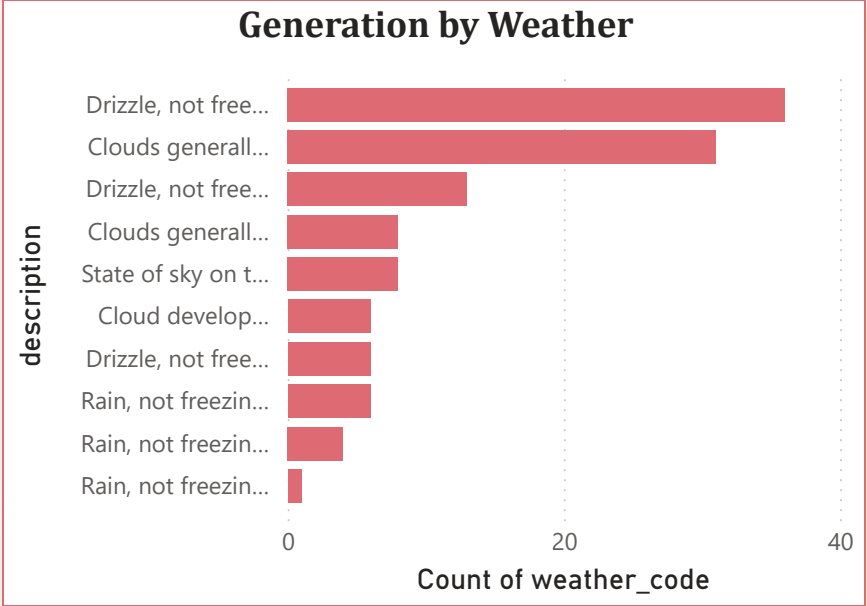


27.17

 Average temperature

4.06K

 Total Consumption



FILTERS

date

01-02-202601-06-2026

weather_code

All

4.01K

Total Generation

1.41

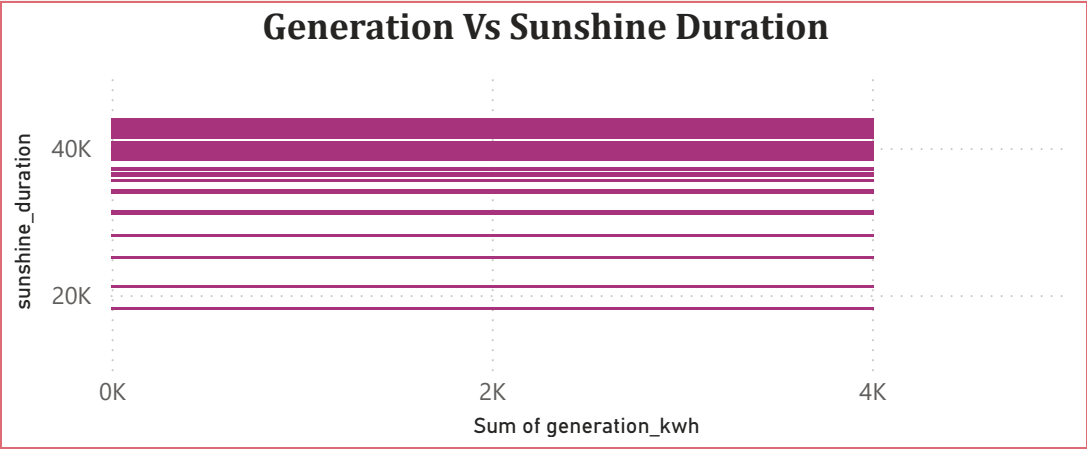
Avg generation

6.60

Max Generation

SOLAR PERFORMANCE

Analyze solar energy generation trends and performance



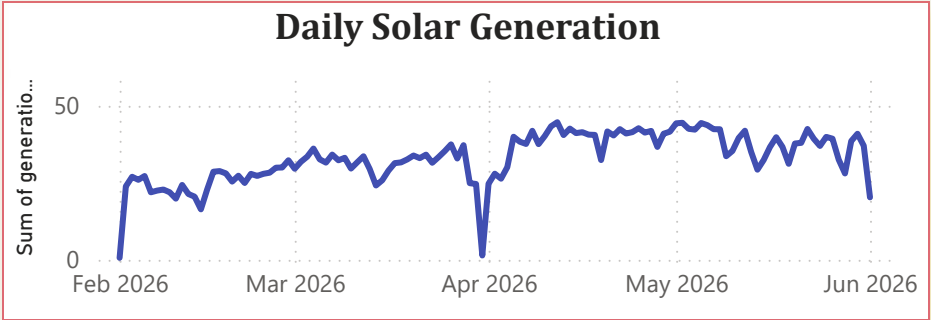
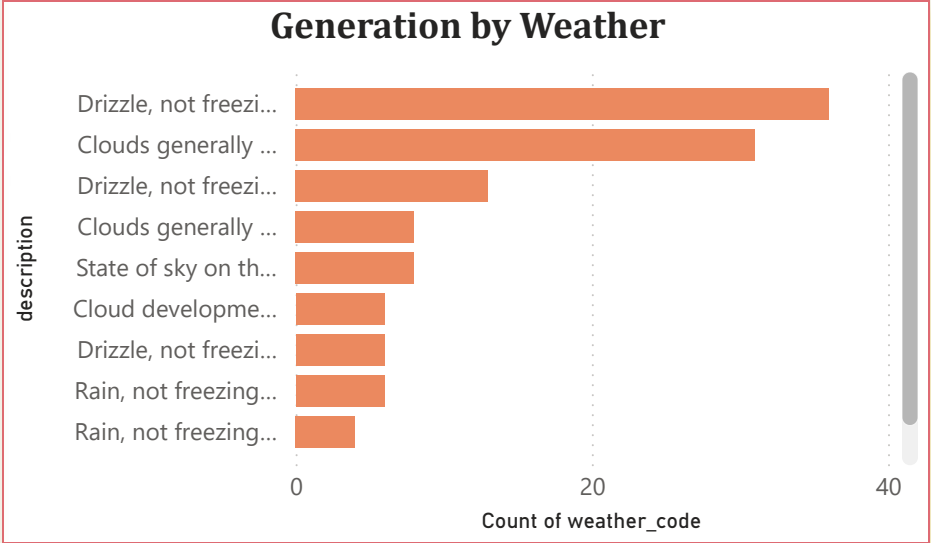
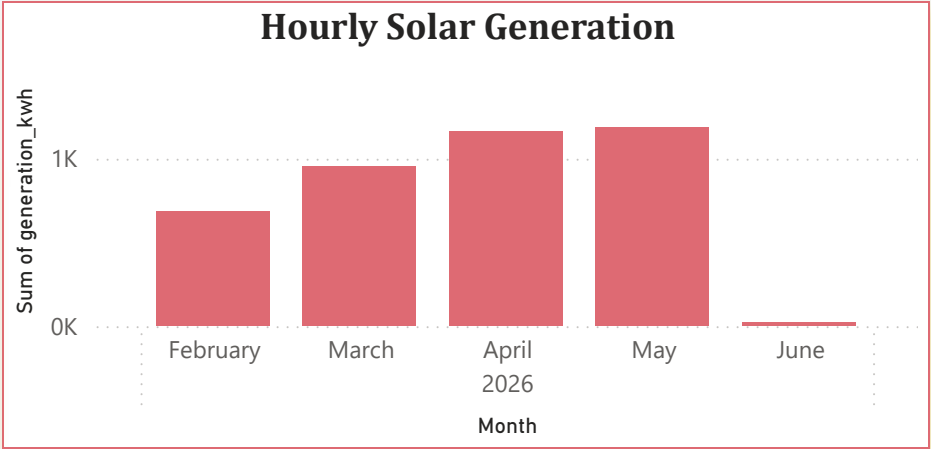
Year	Month	Day	Sum of generation_kwh	Sum of sunshine_duration	Sum of ter
2026	February	1	1.29	49,18,499.61	
2026	February	2	23.98	49,18,499.61	
2026	February	3	26.46	49,18,499.61	
2026	February	4	25.94	49,18,499.61	
2026	February	5	27.26	49,18,499.61	
2026	February	6	22.43	49,18,499.61	
2026	February	7	22.72	49,18,499.61	
2026	February	8	22.23	49,18,499.61	
Total			4,011.43	49,18,499.61	

41.33K

Avg Sunshine Duration

27.17

Average temperature



FILTERS

date

01-02-202601-06-2026

hour_ts

02-02-202601-06-2026

2.00K

Battery charge

1.80K

Battery Discharge

100

Max battery soc

20

Min battery soc

16.50

Average...

Battery SOC

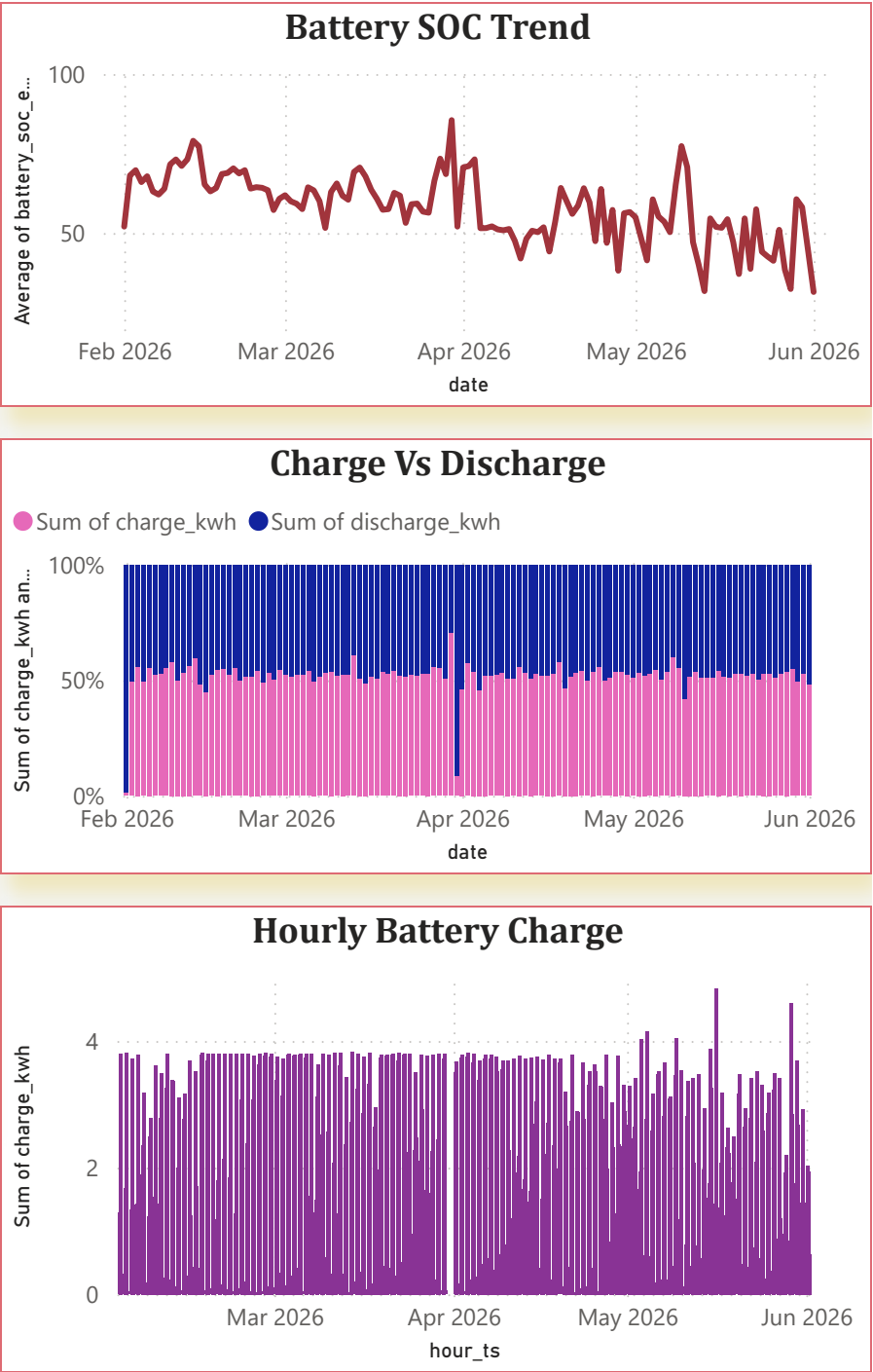
58.15

0.00116.31

Battery Analysis

Analyze battery charging, discharging and state of charge(SOC)

Year	Month	Day	Average of battery_soc_eoh	Sum of charge_kwh	Sum of discharge_kwh
2026	February	1	32.38	0.58	
2026	February	2	71.79	14.06	1
2026	February	3	67.58	16.47	1
2026	February	4	68.29	14.58	1
2026	February	5	66.08	16.85	1
2026	February	6	64.54	14.05	1
2026	February	7	61.63	15.43	1
2026	February	8	62.83	15.87	1
2026	February	9	69.08	15.46	1
2026	February	10	74.08	12.76	1
2026	February	11	71.96	14.24	1
2026	February	12	69.67	14.53	1
2026	February	13	76.33	12.46	
2026	February	14	80.63	9.79	
2026	February	15	72.63	12.82	1
2026	February	16	63.29	17.96	1
2026	February	17	61.88	17.89	1
2026	February	18	66.67	17.15	1
2026	February	19	69.58	15.49	1
2026	February	20	60.28	15.48	1
Total			58.15	1,996.22	1,80



FILTERS

date

01-02-2026

31-05-2026



weather_code

All

152.10

☂ Total Rainfall

42.34

☁ Avg Cloud cover

27.17

🌡 Average temperature



Weather Analysis

Analyze weather conditions and their impact on solar generation

15.02

🌿 Average Wind Speed

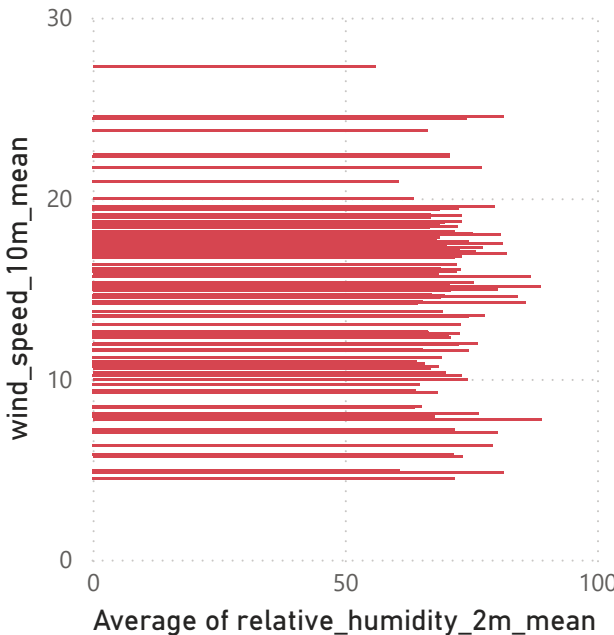
71.12

💧 Average humidity

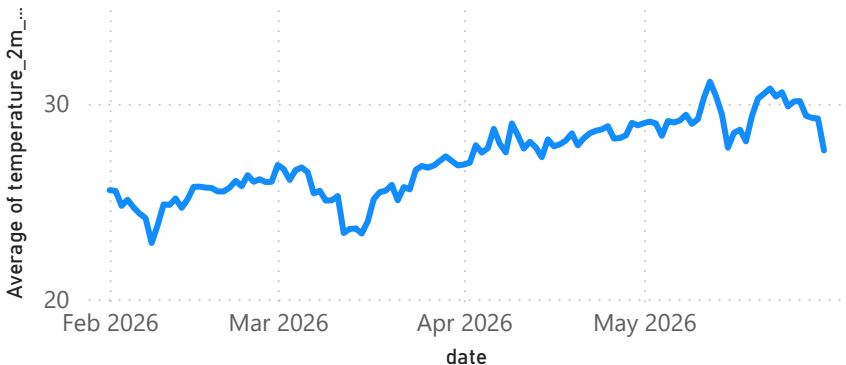
41.33K

☀ Avg Sunshine...

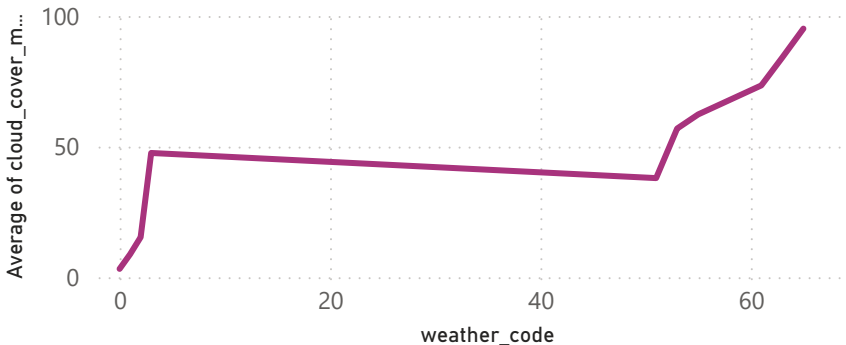
Wind Speed vs Humidity



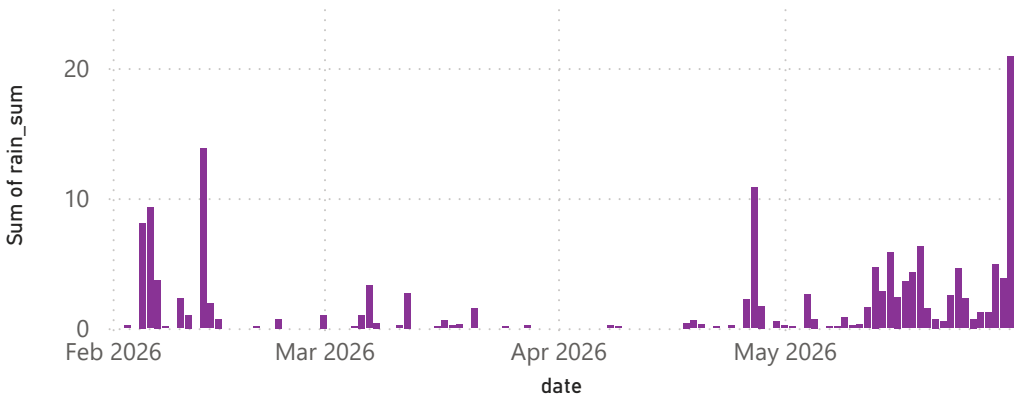
Temperature Trend



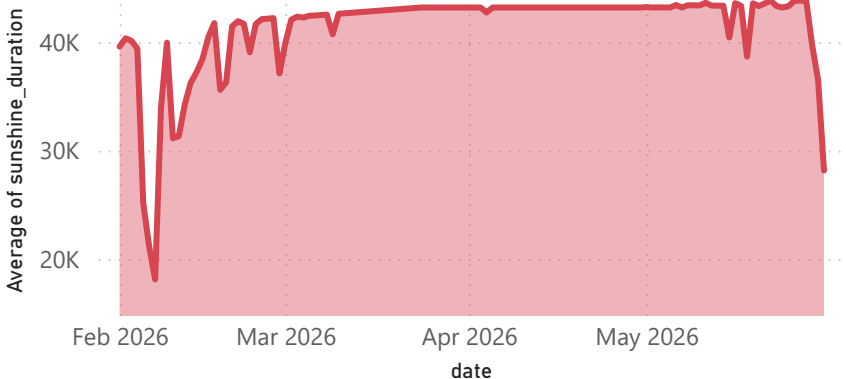
Cloud Cover by Weather Condition



Rainfall Trend



Sunshine Duration Trend



FILTERS

date

01-02-2026

01-06-2026



hour_ts

02-02-2026

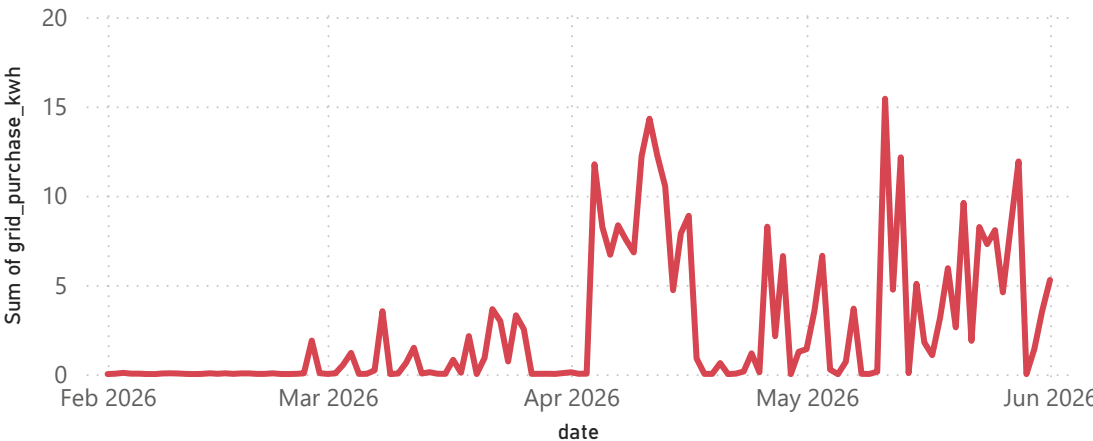
01-06-2026



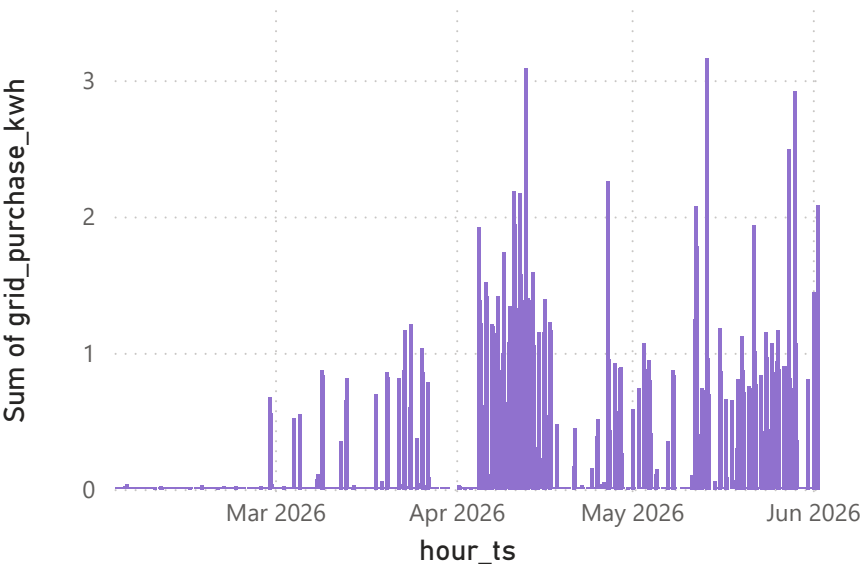
Grid Analysis

Analyze grid energy import, export, and overall grid utilization

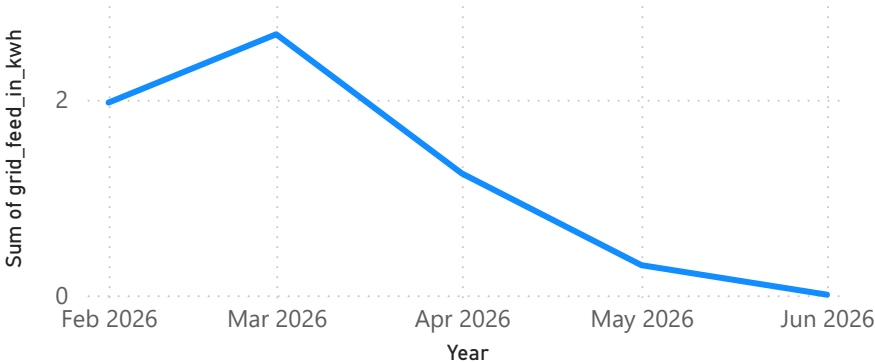
Grid Purchase Trend



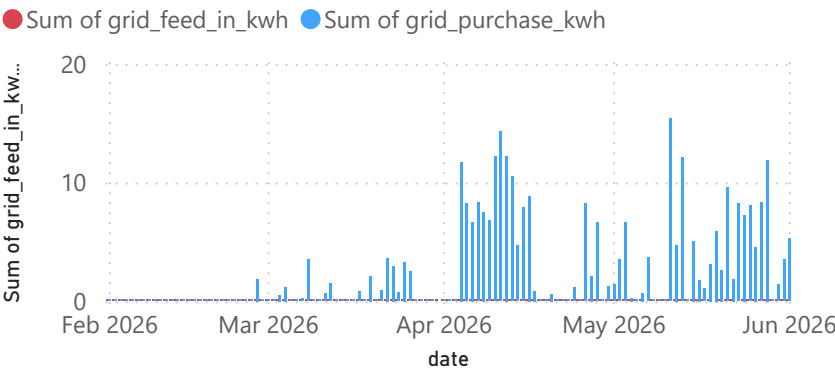
Hourly Grid Purchase



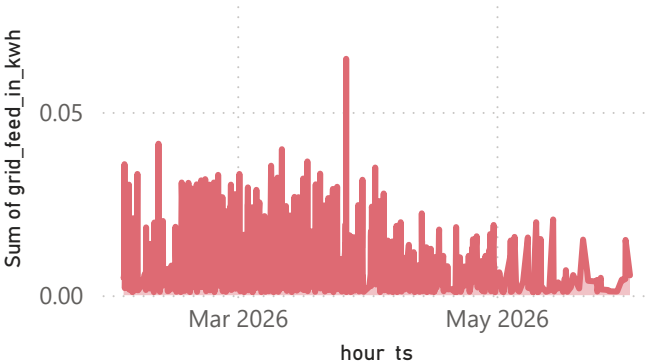
Grid Feed in Trend



Grid Feed In Vs Grid Purchase



Hourly Grid Feed In



6.20

↑ Grid Feed In

307.68

↓ Grid Purchase

0.05

⬆ Avg grid feed in

2.56

⬆ Avg grid purchase

0.16

↑ Max Grid Feed In

FILTERS

date

01-02-202601-06-2026

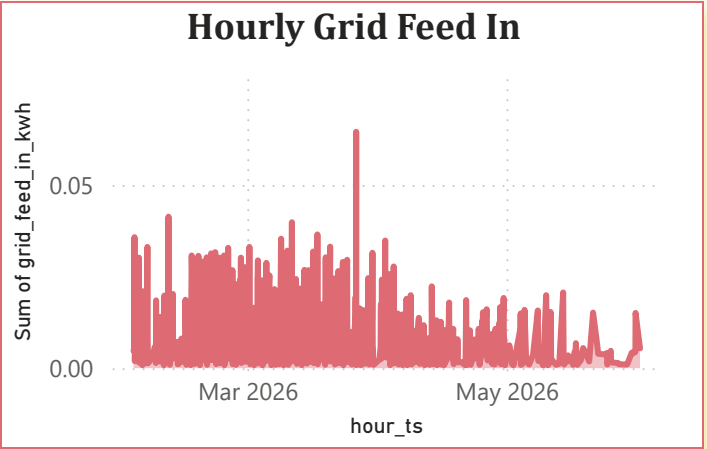
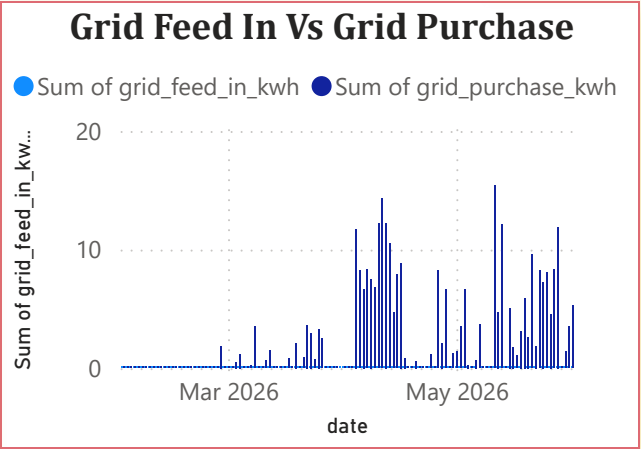
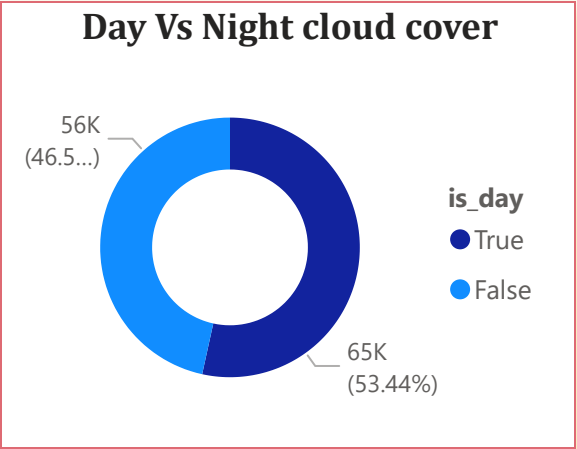
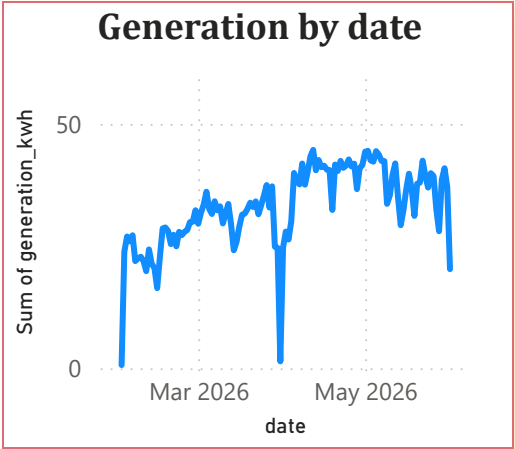
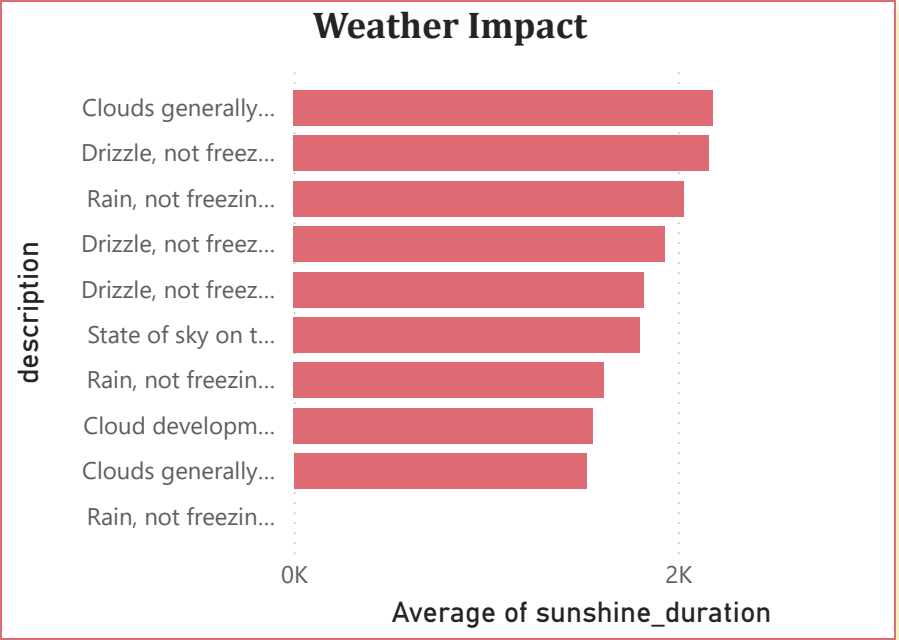
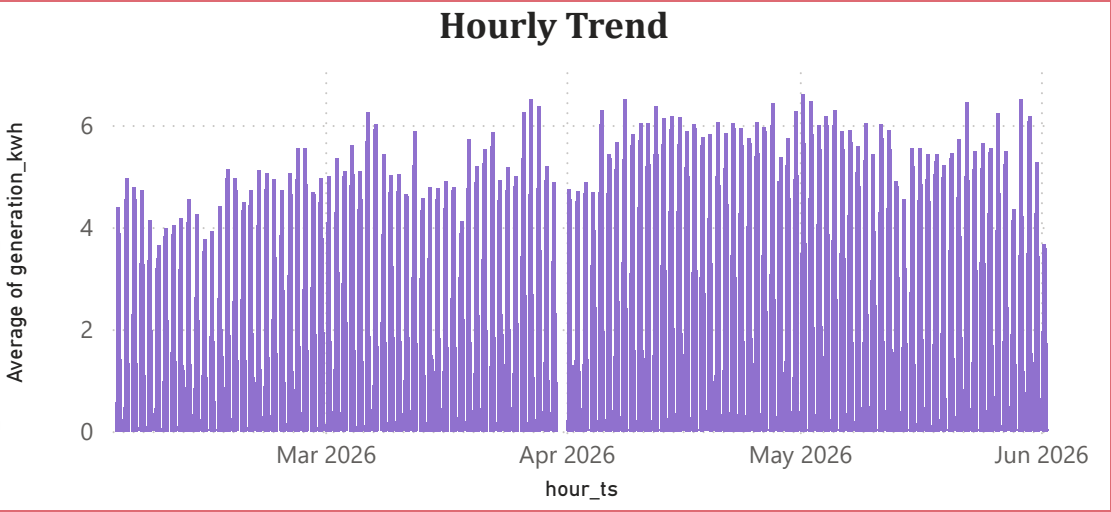
hour_ts

02-02-202601-06-2026



Hourly Analysis

Track generation performance across each hour with weather impact insights



4.01K

 Total Generation

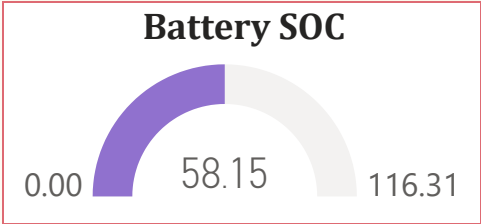
2.855K

 count Hour ts

41.33K

 Avg Sunshine...

Battery SOC



0.00 58.15 116.31

2.00K

 Battery charge

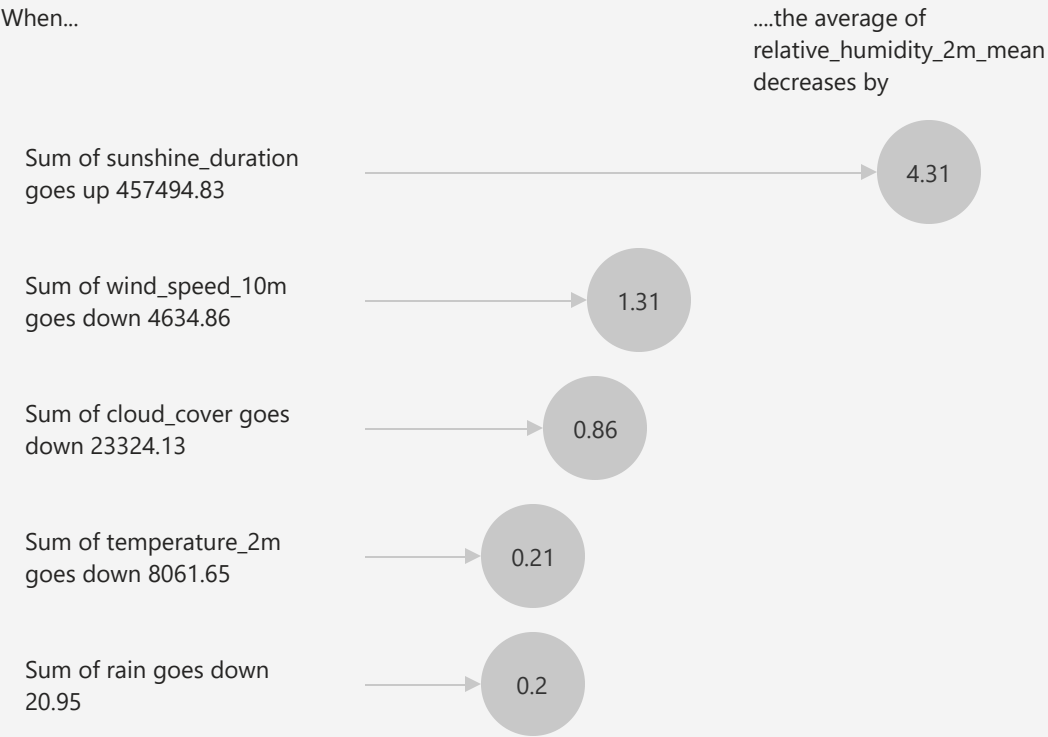
Key influencers Top segments



What influences relative_humidity_2m_mean to

Decrease

 ?



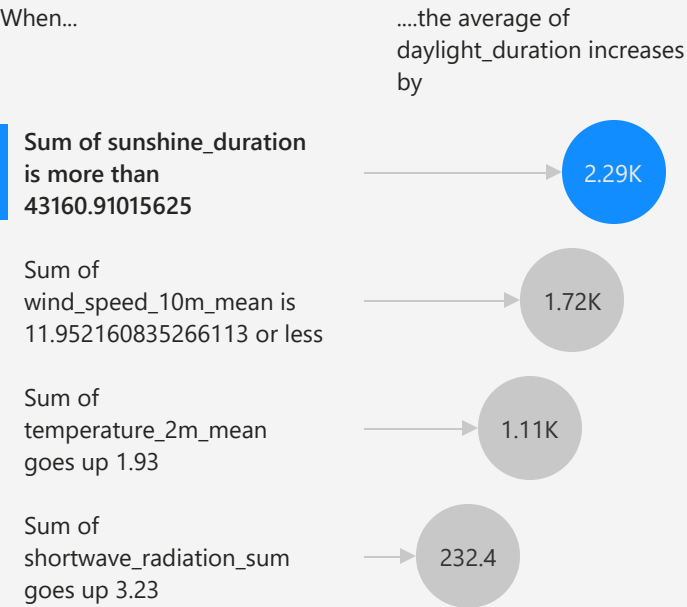
Key influencers Top segments



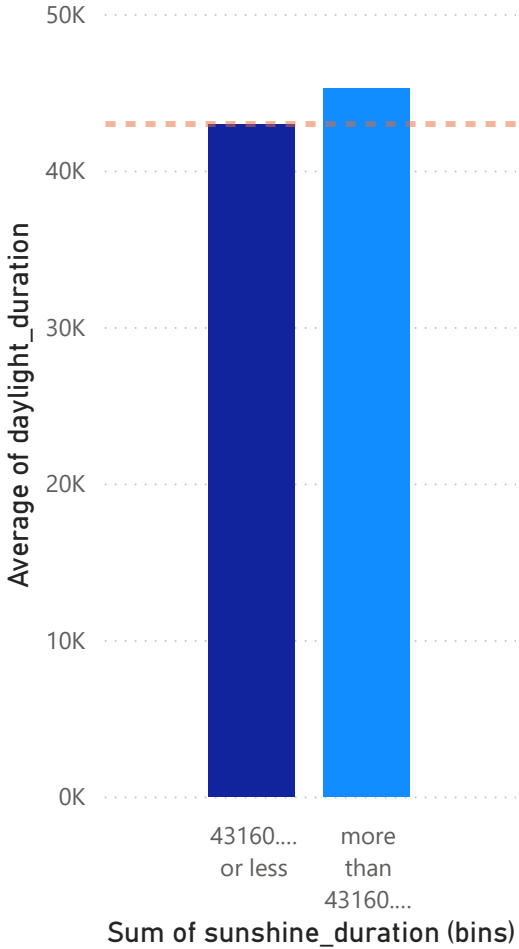
What influences daylight_duration to

Increase

 ?



← daylight_duration is more likely to increase when Sum of sunshine_duration is more than 43160.91015625 than otherwise (on average).



☐ Only show values that are influencers

Solar Generation Analysis Dashboard

Solar ♦ Battery ♦ Weather Analytics

Solar generation is highly weather dependent, with May 2026 showing peak output. March 31 shows a sharp decline. Consumption often matches or exceeds generation, increasing grid dependency. Cloudy and drizzle conditions reduce efficiency. Battery SOC peaks on March 30, while sunshine duration drops lowest on February 7, 2026.